

Arnav Srikanth

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EDUCATION

Carnegie Mellon University, Pittsburgh, PA

Master of Science in Mechanical Engineering, GPA: 4.0/4.0

December 2026

MetaMobility Lab

Relevant Courses: Mechanics of Manipulation, Advanced Mechatronic Design, Machine Learning and Artificial Intelligence for Engineers, Intro to Robot Learning, Bioinspired Robot Design and Experimentation, Computer Vision, Modern Control Theory, Soft Robots

University of Illinois Urbana-Champaign, Champaign, IL

Bachelor of Science in Mechanical Engineering

May 2022

Midwestern Robotics Design Competition, iRobotics Robobrawl, Illini Esports Community Events Chair

SKILLS

Applications: SolidWorks/SolidWorks Simulate, PTC Creo, Fusion360, FEA, EasyEDA (PCB Design)

Manufacturing: 3D printing, CNC/manual milling, laser cutting, rapid prototyping, injection molding, ASME Y14.5

Programming: Python, MATLAB, C/C++

WORK EXPERIENCE

Robotics Mechanical Engineering Intern | ASM, Phoenix, AZ

June 2026 - August 2026

- Engineered automation solution for semiconductor equipment maintenance, replacing a manual showerhead replacement process involving a 56 kg component and reducing field service engineer requirements by 50%
- Designed and rapidly prototyped robotic end effectors and support hardware using PTC Creo, machining, and 3D printing to allow robot to manipulate components of semiconductor tools during preventative maintenance with a ± 3 mm measured margin of error
- Built and validated simulation environments in PyBullet and RoboDK to assess workspace constraints, collision avoidance and trajectory planning using RRT-Connect algorithms for a JAKA Zu18 6 DoF collaborative robot
- Developed computer vision pipeline and encoder-driven closed loop control scheme to integrate COGNEX industrial automation cameras, JAKA Zu18, and Kolver KDucer electric screwdriver using Python to communicate via MODBUS TCP to automate fastener removal

Graduate Student Researcher | MetaMobility Lab, Pittsburgh, PA

August 2025 - Aug 2026

- Overhauled hip exoskeleton mechatronics and developed custom power delivery PCB, reducing total system weight by 10% and eliminating external cabling, resulting in a 30% reduction in subject setup time

Mechanical Designer | Plug Power, Concord, MA

November 2023 - April 2025

- Implemented P&I diagram for a stationary 250kW hydrogen fuel cell system by laying out electronic, thermodynamic and mechanical components to allow for ease of access to reference material for all engineers
- Led cross-disciplinary team of in design review meetings to iterate through solutions for product repackaging
- Overhauled system packaging hardware and piping design in PTC Creo for CNC milling and sheet metal manufacturing to achieve 40% and 35% reduction in overall volume and service costs respectively
- Rapidly burned down risk to accelerate project timeline by quickly iterating through schematics and drawings with Microsoft Visio and PTC Creo

Mechanical Engineer | Smartville Inc, Carlsbad, CA

December 2022 - August 2023

- Created and maintained SolidWorks CAD model and BOM for 1500 kWh battery energy storage system, incorporating DFM and DFA feedback during refinement stage to drive cost reduction efforts
- Produced machine drawings in accordance to ASME Y14.5 GD&T standards to outsource sheet metal fabrication
- Collaborated with cross-disciplinary teams to construct, fabricate, and operate test fixtures for electrical subsystems as well as prototype cable harnesses for high and low voltage components
- Coordinated with APAC and domestic suppliers to source electrical components in compliance with UL 1973/74

Mechanical Designer | FORGE LAB, Champaign, IL

July 2021 - August 2022

- Designed LIDAR-based data collection system in SolidWorks to measure atmospheric composition
- Conducted design reviews with NASA Wallops engineers to implement design and manufacturing changes to address concerns about structural stability under rapid vibrations and large temperature deltas